Elekta Neuromag® Triux

With internal helium recycler

Not for USA

The next leap forward in functional mapping

Elekta

Technical specifications

Elekta Neuromag® Triux with internal helium recycling (art. no. NM25000N) is a comprehensive bioelectromagnetic measurement system for functional brain studies:

- 306-channel neuromagnetometer, which optimally combines both planar gradiometers as well as magnetometers
- Highest available information capture
- Uniquely effective interference elimination technologies with unparalleled robustness even in the presence of intense, nearby sources of interference
- Utmost flexibility in tailoring the system for the particular clinical and/or research needs thanks to a wide range of options
- Groundbreaking zero helium boil-off technology

Elekta Neuromag® Triux with internal helium recycler is not available for sale in all markets.

Sensor array

State of the art sensor array featuring densest spatial sampling and lowest intrinsic noise in the industry:

- Whole-head sensor array with realistic helmet shape
- 1,220 cm^2 sensor coverage
- 102 high-precision triple-sensor elements
- 204 planar gradiometers
- 102 magnetometers
- 35 mm average distance between each sensor element
- 20 mm effective channel separation
- 17.0 mm base length (planar gradiometers)
- <0.1 % geometrical imbalance
- ±20 nT dynamic range
- 18 mm average distance between pick-up coils and room temperature surface
- <1 second detrapping cycle for individual channels with integrated thin-film heaters
- 2.5 minute detrapping cycle for all channels
- Typical white noise levels:

Sensor type	White noise
Gradiometer	3.6 fT/cm/$\sqrt{Hz}$
Magnetometer	3.5 fT/$\sqrt{Hz}$

- Guaranteed gradiometer noise levels (percentage of sensors):

Frequency	Noise[1]
1–10 Hz	<20 (100 %)
	<12 (96 %)
60–70 Hz	<10 (100 %)
	<5 (96 %)

[1] fT/cm/$\sqrt{Hz}$

- Guaranteed magnetometer noise levels (percentage of sensors):

Frequency	Noise[1]
1–10 Hz	<20 (100 %)
	<12 (96 %)
60–70 Hz	<10 (100 %)
	<5 (96 %)

[1] fT/$\sqrt{Hz}$

- <0.1 % residual crosstalk between sensors (corrected)

Data acquisition

- Up to 474 simultaneously sampled channels
- 306 MEG channels
- Up to 128 EEG channels
- 12 bipolar analog input channels for physiological signals
- 12 auxiliary analog input channels (±10 V)
- 16 trigger (TTL) channels
- 1–5 kHz sampling rate, configurable by the user
- 24-bit output
- 2.0 fT/cm least significant bit (gradiometer channels)
- 2.5 fT least significant bit (magnetometer channels)
- MEG high-pass filter cutoff frequencies: DC–10 Hz
- EEG high-pass filter cutoff frequencies: DC, 0.03, 0.1, and 10 Hz

Interference elimination

Proprietary MaxFilter™ and signal space projection technologies for elimination of spatial and spatiotemporal interferences:

- Online signal space projection (SSP) for adaptive elimination of external interferences with up to 40 dB attenuation
- MaxFilter™ technology for offline spatial suppression of interferences with up to 40 dB attenuation for frequencies up to 100 Hz
- MaxFilter™ technology for offline spatiotemporal elimination of even intense artifacts from distant and nearby sources with up to 130 dB attenuation
- Automatic offline detection and exclusion of defective channels

Movement compensation

Patented movement compensation technology for continuous elimination of motion artifacts:

- Continuous head position tracking
- User-defined update frequency up to 1 kHz
- Offline compensation for temporary movements of the patient's head
- Offline transformation of data between different head positions

Gantry
Motorized gantry with integrated connectors for auxiliary electrodes and stimuli:
- Reclined measurement position (60° recline)
- Upright measurement position (68° recline)
- Supine measurement position (0°)
- Button-operated electric lifting motor
- Safety locking latch
- Integrated connectors for EEG and auxiliary channels
- Automatic detection of gantry position
- Weight (empty): 450 kg (992 lb)
- Dimensions (upright): 950 × 1,660 × 2,220 mm (37.4 × 65.4 × 87.4 in)
- Dimensions (supine): 950 × 1,340 × 1,990 mm (37.4 × 52.8 × 78.3 in)

Cryogenics
Cryogenics featuring zero helium boil-off technology and fastest stabilization after change of measurement position:
- 78-liter (20.6 gal) liquid helium dewar
- <50 l/year (<13.2 gal/year) helium loss rate (excluding loss during possible maintenance breaks)
- 365-day refill interval (excluding possible maintenance breaks)
- Minimum 35 hours/week available measurement time
- <48-hour warmup-cooldown cycle
- Safety exhaust line to an outdoor vent
- Automatic scheduling and monitoring of helium recycling and cryogenic system
- Liquid helium transfer siphon
- Liquid helium gauge for storage dewars
- Helium transfer accessories

See NM25211B for more details.

Head phantom
Precision-engineered head phantom with both current and magnetic dipoles for verification of localization accuracy:
- 32 current dipoles
- 4 magnetic dipoles
- 89 mm (3.5 in) radius
- Guaranteed localization accuracy (aggregate): ≤5.0 mm (≤0.2 in)
- Typical localization accuracy (aggregate): ≤2.0 mm (≤0.08 in) for current dipoles

Co-registration
Landmark-based co-registration of patient's anatomy with magnetic resonance images:
- Co-registration using anatomical landmarks
- Digitization of additional surface points for visual verification

Acquisition workstation
Powerful workstation for data acquisition and post-processing of measurements:
- High-end graphical 6-core workstation
- Linux operating system
- 16 GB RAM
- 1 TB SCSI hard disk
- CD/DVD+RW writer
- 30" TFT monitor
- Real-time visualization of raw data for user-selected channels
- Recording of raw data
- Programmable stimulation sequences and timing
- Online averaging of 1–32 categories
- Automatic rejection of signal epochs containing artifacts
- Visualization of online averages
- Noise level monitoring
- Automatic and manual noise optimization
- Loadable acquisition presets
- Post-acquisition interference elimination

Operating environment
Required operating environment within the magnetically shielded room:
- Air temperature: 20–28 °C (68–82 °F)
- Minimum air flow: 30 l/s (8 gal/s)
- Relative humidity: 40–70 %

Required operating environment within the equipment and operator area:
- Air temperature: 20–24 °C (68–75 °F)
- Relative humidity: 40–70 % (non-condensing)

Power
Requires both single-phase and three-phase power:
- Single-phase input voltage: 100, 115, 200, 230, or 240 V (47–63 Hz)
- Three-phase input voltage: 200 V (50/60 Hz), 380–415 V (50 Hz), or 480 V (60 Hz)
- Single-phase power consumption: 5,700 W (average)
- Three-phase power consumption: 5,000 W (average)

Built-to-order options

32-channel EEG
Art. no. NM23904N

Integrated EEG subsystem with 32 channels:

- 32 unipolar EEG channels
- <0.4 µVrms (0.5–100 Hz) noise at 10 kΩ impedance
- 100 MΩ input impedance at DC
- Software-controllable gain for individual channels (640, 2,000, or 20,000)
- 100 dB common-mode rejection ratio (with active ground)
- Fixed analog high-pass filters: 0 Hz (DC), 0.03 Hz, 0.1 Hz, and 10 Hz

64-channel EEG
Art. no. NM23889N

Integrated EEG subsystem with 64 channels:

- 64 unipolar EEG channels
- <0.4 µVrms (0.5–100 Hz) noise at 10 kΩ impedance
- 100 MΩ input impedance at DC
- Software-controllable gain for individual channels (640, 2,000, or 20,000)
- 100 dB common-mode rejection ratio (with active ground)
- Fixed analog high-pass filters: 0 Hz (DC), 0.03 Hz, 0.1 Hz, and 10 Hz

128-channel EEG
Art. no. NM23893N

Integrated EEG subsystem with 128 channels:

- 128 unipolar EEG channels
- <0.4 µVrms (0.5–100 Hz) noise at 10 kΩ impedance
- 100 MΩ input impedance at DC
- Software-controllable gain for individual channels (640, 2,000, or 20,000)
- 100 dB common-mode rejection ratio (with active ground)
- Fixed analog high-pass filters: 0 Hz (DC), 0.03 Hz, 0.1 Hz, and 10 Hz

Patient positioning

Patient couch
Art. no. NM23125N

Roll-away patient couch on wheels:
- Moveable upper bed
- Transparent, removable side guards
- Safety belt
- Detachable head support
- Weight: 75 kg (165 lb)
- Dimensions: 2,220 × 740 × 760 mm (87.4 × 29.1 × 29.9 in)
- Maximum load: 135 kg (298 lb)

Patient chair
Art. no. NM23124N-A

Roll-away patient chair on wheels:
- Adjustable height
- 2 removable table plates, non-slip table top (separate washable silicon matt)
- Weight: 80 kg (176 lb)
- Dimensions: 1,988 (with handle) × 650 × 784 mm (78.3 × 25.6 × 30.9 in)
- Maximum load: 135 kg (298 lb)

Patient chair options

Chair fine-adjustment system*
Art. no. NM25745N

System for fine-adjusting horizontal position and backrest recline of the patient chair:
- Non-magnetic operating console attached to the gantry
- Operating levers for smooth and consistent adjustments
- Horizontal movement system on the floor
- Console dimensions: 346 x 208 x 653 mm (13.6 x 8.2 x 25.7)
- Requires the patient chair (NM23124N-A)

*Patents pending

Pediatric comfort set
Art. no. NM25744N

Non-magnetic chair cushioning for pediatric patients and research participants:
- Set of cushions: 2 seats, 2 pairs of armrests, and 2 saddle pieces
- Adjustable footrests
- Requires the patient chair (NM23124N-A)

Accessories

Stimulus presentation workstation
Art. no. NM21711N

Stimulus presentation system:
- Stimulus generation software
- Dedicated stimulation workstation
- TTL-based synchronization with data acquisition

High-fidelity visual stimulator
Art. no. NM24034N

High-fidelity video projection system:
- Native 16:9 three-panel DLP video projector
- Symmetric black-to-white and white-to-black transition time
- 26 ms delay from trigger presentation
- <1 ms jitter (frame-to-frame) at 60 Hz refresh rate
- 1,400 × 1,050 pixel native resolution
- 16-bit color depth
- 24–120 Hz refresh rate
- 112 cm (44 in) standalone back-projection screen on wheels

Auditory stimulator
Art. no. NM24035N

Stereophonic auditory stimulus delivery system:
- Non-magnetic tubal-insert earphone set
- Independent delivery of auditory stimuli to each ear
- >80 dB sound pressure level (SPL)
- >60 dB channel separation
- <1 ms jitter between trigger and stimulus onset
- 50 disposable foam eartips

Somatosensory stimulator
Art. no. NM21709N

Somatosensory stimulus delivery system:
- One electric somatosensory stimulator with controllable amplitude
- 0–99.9 mA output current
- 50, 100, 200, 500, 1,000, or 2,000 µs pulse duration
- Triggered via the acquisition workstation or from within the stimulus presentation workstation

Response device
Art. no. NM20999N

Non-magnetic, single-finger optical response pad:
- Finger-press mode
- Finger-lift mode
- Trigger output
- <1 ms delay between response and recorded event

Video monitoring system
Art. no. NM21712N

A CCTV video monitoring system.
- CCTV video camera
- Monitor
- Cabling

Bidirectional intercom system
Art. no. NM21632N

A bidirectional intercom system for two-way communication between the patient, operator, and nurse.
- Intercom system with half-duplex and simplex modes
- Microphone
- Cabling

Acquisition system UPS
Art. no. NM24062N

An uninterruptible power supply unit for maintaining the Elekta Neuromag® Triux with helium recycling data acquisition system operational during power outage or voltage fluctuation.
- A 3,000 VA uninterruptible power supply for the data acquisition electronics
- Cabling

Workstation UPS
Art. no. NM24063N

An uninterruptible power supply unit for maintaining a workstation operational during power outage or voltage fluctuation.
- A 1,000 VA uninterruptible power supply for a single workstation and peripherals
- Cabling

Consumables

Head position coils

- Head position coil set with five coils in a ready-to-use assembly (art, no. NM23880N)

32-channel EEG caps

- Small (size 46–52 cm) 32-channel EEG cap (art. no. NM23905N)
- Medium (size 52–58 cm) 32-channel EEG cap (art. no. NM23906N)
- Large (size 58–64 cm) 32-channel EEG cap (art. no. NM23907N)

64-channel EEG caps

- Small (size 46–52 cm) 64-channel EEG cap (art. no. NM23890N)
- Medium (size 52–58 cm) 64-channel EEG cap (art. no. NM23891N)
- Large (size 58–64 cm) 64-channel EEG cap (art. no. NM23892N)

128-channel EEG caps

- Medium (size 52–58 cm) 128-channel EEG cap (art. no. NM23895N)
- Large (size 58–64 cm) 128-channel EEG cap (art. no. NM23896N)

Elekta is a global human care company pioneering signicant innovations and clinical solutions for treating cancer and brain disorders. Elekta provides intelligent and resource-efficient technologies that improve, prolong and save patient lives.

www.elekta.com

Focus where it matters

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